Let $(M_k, \omega_k) \xrightarrow{GH} (Y, d_Y)$

$L_k \to M_k$ have curv. $-\sqrt{-1}\,\omega_k$

$|Ric(\omega_k)| \le 1$, $\quad Vol(B(q,1)) > v \quad \forall q \in M_k$

$Vol(M_k, \omega_k) < V \quad \forall k$

Replacing L_k by L_k^m, can assume

$\Phi_{L_k^j} : M_k \to \mathbb{CP}^{N_j}$ defined $\forall k, j > 0$

have uniform Lipschitz bound,

converge to $\Phi_{L_Y^j} : Y \to \mathbb{CP}^{N_j}$

using sections $H^0(Y, L_Y^j) = \left\{ \begin{array}{l} \text{holo. sec. } s \text{ of} \\ L_Y^j \text{ on } Y_{reg} \text{ s.t. } \|s\|_{L^2} < \infty \end{array} \right\}$

Let $W_j = \Phi_{L_Y^j}(Y)$ $\leftarrow$ complex variety

(closure of $\Phi_{L_Y^j}(Y_{reg})$
or limit of $\Phi_{L_k^j}(M_k)$)

We have morphisms

$$f_{j,\lambda} : W_{\lambda j} \to W_j \quad \forall \lambda, j > 0$$

<u>Goal</u> for large j maps $f_{j,\lambda}$ are isomorphisms

i.e. $\Phi_{L_Y^j}$ is a bijection ($\Rightarrow$ homeomorphism)

(see Donaldson-Sun : GH limits of Kähler mflds
& alg. geom)

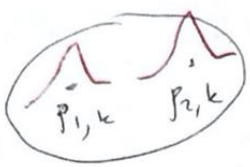
<u>Separating points</u> Given $K > 0$, $\exists \lambda$ s.t. if $p_1, p_2 \in Y$ s.t. $d(p_1, p_2) > K$ then

$$\Phi_{L_Y^\lambda}(p_1) \neq \Phi_{L_Y^\lambda}(p_2)$$

<u>Proof</u> Let $p_{1,k}, p_{2,k} \in M_k$ s.t. $p_{1,k} \to p_1$, $p_{2,k} \to p_2$

From main existence result can find sections $s_{1,k}, s_{2,k}$ of L_k^m for large k s.t. $|s_{i,k}(p_{i,k})| > \frac{1}{2}$, $|s_{i,k}(p_{j,k})| < \frac{1}{100}$, $i \neq j$

$$\|s_{i,k}\|_{L^2} < C.$$



Can pass $s_{i,k}$ to the limit space Y. $\square$

<u>Separating tangents on Y_{reg}</u>: If $K \subset Y_{reg}$ is compact, $\exists \lambda(K)$ s.t. for $\lambda > \lambda(K)$ if $p \in K$, and $v \in T_p Y_{reg}$, $\exists s \in H^0(Y, L_Y^\lambda)$ s.t. $s(p) = 0$, $\nabla_v s(p) \neq 0$.

<u>Proof</u> Let $p_k \to p$, $p_k \in M_k$. Choosing $\lambda(K)$ large and replacing L_k by L_k^λ for $\lambda > \lambda(K)$ can assume

$$d_{GH}\left(B(p_k, \varepsilon^{-1}), \underset{\mathbb{C}^n}{\underset{\uparrow}{B(0, \varepsilon^{-1})}}\right) < \varepsilon$$

For $R>0$ we have holo. coordinates $z_1, \dots, z_n$ on $B(p_k, R)$ & section s of L_k s.t. $\left| |s| - e^{-\frac{1}{2} d(p_k, \cdot)^2} \right| < \delta$

Use argument from before (cutoff & L^2-est.) applied to sections

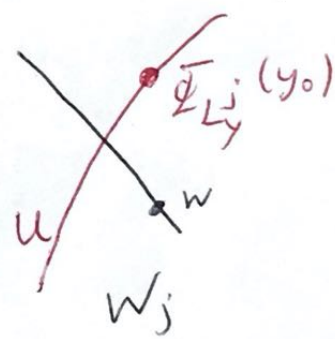
$$z_1 s, \dots, z_n s$$

$\rightsquigarrow$ we get sections $s_1, \dots, s_n$ of L_k^λ s.t. on $B(p_k, 1)$ $\left\| \frac{s_i}{s} - z_i \right\|_{C^{1,\alpha}} < \Psi(\delta, R^{-1})$ $\square$

<u>Lemma</u> For all j, variety $W_j \subset \mathbb{CP}^{N_j}$ is irreducible.

<u>Proof</u> Suppose not. Let $y_0 \in Y_{reg}$ s.t. $\Phi_{L_Y^j}(y_0)$ lies on unique irred. comp U

Let $w \in W_j$ lie on a different component



$\exists$ section P of $\mathcal{O}_{\mathbb{CP}^{N_j}}(\lambda)$ s.t.

$P|_U = 0$, $P(w) \neq 0$

$\rightsquigarrow$ sections s_k of $L_k^{\lambda_j}$

s_k converge to a section s of $L_Y^{\lambda_j}$ vanishing on a neighborhood of $y_0 \in Y_{reg}$.

by unique continuation & since Y_{reg} is connected, $s|_{Y_{reg}} = 0$. Y_{reg} is dense, so $s = 0$ on Y. This contradicts $P(w) \neq 0$ $\square$

<u>Lemma</u> For large j, $\Phi_{L_Y^j} : Y \to W_j$ has finite fibers.

<u>Proof</u> By separating points & tangents results we can assume that $\exists\, U \subset Y_{reg}$ open s.t.

- $\Phi_{L_Y^j}|_U$ is an embedding
- for $w \in \Phi_{L_Y^j}(U)$, $\Phi_{L_Y^j}^{-1}(w) = \{w\}$

Replacing L_Y by a power we can assume this holds for all j.

$$\begin{tikzcd} & Y \arrow[dl, "\Phi_{L_Y^{\lambda j}}"'] \arrow[d, "\Phi_{L_Y^j}"] \\ W_{\lambda j} \arrow[r] & W_j \end{tikzcd}$$

$\Longrightarrow$ maps $f_{j,\lambda} : W_{\lambda j} \to W_j$ are generically one-to-one $\forall\, \lambda, j > 0$.

By separating points, for fixed $r > 0$, we can assume for $w \in W_j$, $\Phi_{L_Y^j}^{-1}(w)$ is contained in an r-ball B

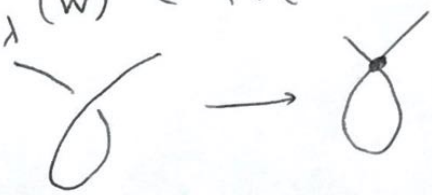
& we have a section $s \in H^0(Y, L_Y^j)$ s.t. $s|_B \neq 0$

It follows that $f_{j,\lambda}^{-1}(w) \subset$ affine subset $\{s^\lambda \neq 0\}$

$\Longrightarrow$ $f_{j,\lambda}^{-1}(w)$ is a finite set
(compact subvariety of $\mathbb{C}^N$)

Let $N(w)$ = number of local irred. components of W_j at w

$\Longrightarrow$ $\forall \lambda$, $\# f_{j,\lambda}^{-1}(w) \leq N(w)$



$\Longrightarrow$ $\# \Phi_{L_Y^j}^{-1}(w) \leq N(w)$ $\square$